RESUME

Basic Information



Name: Liyun Zhang
Nationality: China-Xi'an
Institution: Aonishi Lab, The University of Tokyo
https://aonishilab.jp/
Status: Specially Appointed Assistant Professor
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Research Interests

Embodied AI, Multimodal Learning, Affective Computing, AI for Science.

Education

- **The University of Osaka** 2020.10 - 2024.3
  Ph.D., Information Systems Engineering
  Research focus: Computer vision for image translation / generation / recognition; Robotic perception, SLAM; Multimodal Large Language Model; Embodied AI.
- **Xi'an University of Science and Technology** 2012.9 - 2015.7
  Master of Science, Computer Technology Engineering
  Research focus: Uneven illumination image segmentation and object recognition, Linux-based embedded automation robotic system

Employment

- **Specially-Appointed Assistant Professor** 2025.10 – Current
  Aonishi Lab https://aonishilab.jp/
  The University of Tokyo
  Description: Multimodal Large Language Model, Explainable AI.
- **Specially-Appointed Researcher/Fellow** 2024.4 – 2025.9
  Intelligence and Sensing Lab (ISLab) https://www.is.ids.osaka-u.ac.jp/en/
  The University of Osaka
  Description: Multimodal Large Language Model (MLLMs), Robotic Learning, Multi-annotator learning, Emotion Recognition.
- **Visiting Scholar** 2023.2 - 2024.3
  College of Computing https://animesh.garg.tech/
  Georgia Institute of Technology
  Description: Multi-modal reasoning and LLMs-based embodied AI
- **Research Assistant** 2023.7 - 2024.3
  Graduate School of IST https://www.ist.osaka-u.ac.jp/english/
  The University of Osaka
  Description: Embodied AI and Multi-modal Reasoning
- **Specially Appointed Researcher** 2022.5 - 2023.3
  System Technologies Laboratory
  Sysmex Corporation https://www.sysmex.co.jp/en/index.html
  Description: Identify the area with ointment applied on the forearm (3D partial human body mesh and pose estimation from monocular image)
- **Specially Appointed Researcher** 2021.5 - 2022.3
  Data Science Research Group, CRL
  Sysmex Corporation https://www.sysmex.co.jp/en/index.html
  Description: Time series missing values imputation using GANs-based bidirectional recurrent model on ICU MIMIC-III datasets
- **Research Assistant & Teaching Assistant** 2020.10 - 2021.4
  Cybermedia Center & Graduate School of ES
  The University of Osaka https://www.cmc.osaka-u.ac.jp/?lang=en
  Description: Mainly worked on Image translation / generation, SLAM and intelligent

robot research & assisting graduate students in experiments.

- **Embedded Software Engineer** 2017.12 - 2018.10
  Intelligent Terminal Software Group, Xi'an Research Institute
  ZTE Corporation https://www.zte.com.cn/global/index.html
  Description: Research and development of vehicle audio and power software
- **Software R&D Engineer** 2016.11 - 2017.3
  Software R&D Group, Wuhan Research Institute
  Huawei https://www.huawei.com/us/
  Description: Power management development of smartphone on MTK platform
- **Embedded Software Engineer** 2015.7 - 2017.12
  Software R&D Group, Xi'an Research Institute
  Huaqin Technology https://en.huaqin.com/
  Description: Smartphone software development, image recognition and robot vision algorithm development

Publications († Corresponding author; * Equal contribution)

(A) Journal Articles (Peer-reviewed)

● Xuanmeng Sha, Tomohiro Mashita, Naoya Chiba, **Liyun Zhang**. 3D Local Feature Learning and Analysis on Point Cloud Parts via Momentum Contrast. Sensors, 26(3), 1007. (**Sensors**).

● **Liyun Zhang**, Photchara Ratsamee, Zhaojie Luo, Yuki Uranishi, Manabu Higashida, Haruo Takemura. Panoptic-Level Image-to-Image Translation for Object Recognition and Visual Odometry Enhancement. 2023 IEEE Transactions on Circuits and Systems for Video Technology, 34 (2), 938-954. (**TCSVT**).

● **Liyun Zhang**, Nanyan Liu, Yuanbin Hou, Xiaojian Liu. Uneven Illumination Image Segmentation Based on Multi-threshold S-F [J]. 2014 Opto-Electronic Engineering, 41 (7), 81-87. (**OEE**).

(B) Conference Papers (Peer-reviewed)

● Xuanmeng Sha, **Liyun Zhang**†, Tomohiro Mashita, Naoya Chiba, Yuki Uranishi. 3DGesPolicy: Phoneme-Aware Holistic Co-Speech Gesture Generation Based on Diffusion Policy. 2026 IEEE-RAS International Conference on Humanoid Robots (**HUMANOIDS**).

● **Liyun Zhang**, Xuanmeng Sha, Fengkai Liu, Shuqiong Wu. AcoustEmo: An Utterance-Aware Acoustic Q-Former for Open-Vocabulary Emotion Reasoning. 2026 Conference of the International Speech Communication Association, 2026, pp. 7447–7451, doi: 10.21437/Interspeech.2026-2364 (**INTERSPEECH**).

● **Liyun Zhang**, Mamada Naoya, Kentaro Sakai, Takeo Hoshi, Toru Aonishi. Neural-Net work-based Probabilistic Surrogate Modeling for Rapid Bayesian Inference in Thomson Scattering Diagnostics. 2026 In the 7th International Conference on Data-Driven Plasma Science (**ICDDPS**).

● Mamada Naoya, Masayuki Tokitani, **Liyun Zhang**, Toru Aonishi, Takeo Hoshi. Deep Learning-Based Automated Detection of Clustered Helium Bubbles in Ion-Irradiated Materi als. 2026 The 9th International Conference on Matter and Radiation at Extremes (**ICMRE**).

● Xuanmeng Sha, **Liyun Zhang**†, Tomohiro Mashita, Naoya Chiba, Yuki Uranishi. Pho neme-Aligned Holistic Gesture Synthesis via Robotics-Inspired Control. 2026 In the 29th Meeting on Image Recognition and Understanding (**MIRU**).

● Xuanmeng Sha, **Liyun Zhang**†, Tomohiro Mashita, Naoya Chiba, Yuki Uranishi. 3DFacePolicy: Speech-Driven 3D Facial Animation Based on Diffusion Policy. 2026 IEEE International Conference on Robotics and Automation, IEEE, 2026. pp. 9727-9734 (**ICRA**).

● **Liyun Zhang**, Zheng Lian, Hong Liu, Takanori Takebe, Yuta Nakashima. SimLabel: Similarity-Weighted Semi-supervision for Multi-annotator Learning with Missing Labels. 2026 The 40th Annual AAAI Conference on Artificial Intelligence, Vol. 40, No. 33, pp. 28328-28336. (**AAAI**).

● **Liyun Zhang**, Zheng Lian, Hong Liu, Takanori Takebe, Shozo Nishii, Yuta Nakashima. QuMAB: Query-based Multi-annotator Behavior Pattern Learning. 2026 The 40th Annual AAAI Conference on Artificial Intelligence, Vol. 40, No. 33, pp. 28319-28327. (**AAAI**), (**Oral**).

● Xuanmeng Sha, **Liyun Zhang**†, Tomohiro Mashita, Naoya Chiba, Yuki Uranishi. Robotics-inspired Control for Audio-driven 3D Facial Motion Synthesis. 2025 In the 28th Meeting on

Image Recognition and Understanding (**MIRU**).

● **Liyun Zhang**, Zhaojie Luo, Shuqiong Wu, Yuta Nakashima. MicroEmo: Time-Sensitive Multimodal Emotion Recognition with Subtle Clue Dynamics in Video Dialogues. 2024 In Proceedings of the 2nd International Workshop on Multimodal and Responsible Affective Computing. 2024. pp. 110-115. (**MRAC'24 @ ACMMM**).

● **Liyun Zhang**, Photchara Ratsamee, Bowen Wang, Zhaojie Luo, Yuki Uranishi, Manabu Higashida, Haruo Takemura. Panoptic-aware Image-to-Image Translation. 2023 IEEE/CVF Winter Conference on Applications of Computer Vision, IEEE, 2023. pp. 259-268. (**WACV**).

● **Liyun Zhang**, Photchara Ratsamee, Yuki Uranishi, Manabu Higashida, Haruo Takemura. Thermal-to-Color Image Translation for Enhancing Visual Odometry of Thermal Vision. 2022 IEEE International Symposium on Safety, Security, and Rescue Robotics, IEEE, 2022. pp. 33-40 (**SSRR**).

(C) Conference Papers (In-review)

● **Liyun Zhang**, Xuanmeng Sha, Fengkai Liu, Shuqiong Wu, Bowen Wang, Hong Liu, Zheng Lian. A Unified Evaluation Framework for Individual Tendency Learning in Multi-Annotator Scenarios. arXiv preprint (in review **CVPR** 2027).

● Fengkai Liu, Hao Su, Haozhuang Chi, Rui Geng, Congzhi Ren, Xuqing Liu, Chenfei XU, Yuichi Ohsita, **Liyun Zhang**†. Event-Driven Proactive Assistive Manipulation with Grounded Vision-Language Planning. arXiv preprint arXiv:2603.23950 (in review **ICRA** 2027).

● **Liyun Zhang**, Mamada Naoya, Kentaro Sakai, Takeo Hoshi, Toru Aonishi. Heteroscedastic Neural Surrogate Modeling for Robust and Rapid Bayesian Inference in Fusion Plasma Diagnostics. arXiv preprint (in review **CVPR** 2027).

(D) Journal Articles (In-review)

● Jiachen Luo, Qu Yang, Jiajun he, Yining Hua, Zheng Lian, Yuanchao Li, Siyang Song, Wen Wu, Dingdong Wang, Shuai Shen, Jingyao Wu, Guimin Hu, He Hu, Yong Li, Zixing Zhang, Jiadong Wang, Sifan Zhou, Zuojin Tang, Canran Xiao, Sheng Xu, Zhenjun Zhao, Xiangyang Xue, Sicheng Zhao, Yong Dai, Tomoki Toda, Licai Sun, Kailai Yang, **Liyun Zhang**, Cong Cai, Jiamin Du, Ziyang Ma, Mingjie Chen, Chengxuan Qian, Zhenlong Yuan, Xie Chen, Huy Phan, Lin Wang, Bjoern Schuller, Joshua Reiss. A Comprehensive Review in Unimodal and Multimodal Emotion Recognition. PsyArXiv preprint (in review **ACM Computing Surveys** 2026).

Awards

- **Special Contribution Award** 2017.3
  Solved the problem of smartphone battery level jump in Huawei
- **Star Staff Award** 2016.10
  Acquired "Star Staff" in Huaqin Technology
- **Technology Innovation Award** 2016.3 & 2015.11
  Innovation Second Award 2 times in Huaqin Technology
- **Software copyright** 2015.4
  Steel pipe identification and counting software system
- **Electronic Design Competition Award** 2014.6
  'Automatic orifice positioning system based on embedded Linux' Electronic design competition Third Award
- **Software copyright** 2013.11
  Mine blast hole automatic positioning software system
- **RoboCup Award** 2012.11
  RoboCup China 2012 Middle Size Robot League First Award
- **Excellent Graduation Project (Thesis)** 2012.7
  "Design of Intelligent Bus Stop Announcement System Based on GPS" won the Excellence Award in the Automation Excellent Graduation Project Competition

Scholarships & External Fundings

- 2023 Research Abroad Grant (Osaka University Future Fund Globalization Promotion)
- 2023 Osaka University Graduate School of Information Science Search Assistant
- 2022 Sysmex Student Researcher Program Specially Appointed Researcher S
- 2021 Sysmex Student Researcher Program Specially Appointed Researcher S
- 2020 Osaka University Graduate School of Information Science Search Assistant

Grants & Research Projects

- 2027-2030, Research on Long-Term Affective Interaction Technologies Driven by Robust Subtle Emotion Perception, National Natural Science Foundation of China (NSFC), General Program, Budget Amount (CNY 500,000). **--- Co-Investigator**
- 2026-2027, Event-Driven Epistemic Copilot for Science: Privacy-Preserving Multimodal Human-AI Context Alignment, MEXT (Ministry of Education, Culture, Sports, Science and Technology), Supporting Pioneering Research through AI for Discovery challenges (SPReAD), Budget Amount (JPY 3,900,000). **--- Principal Investigator**
- 2025-2026, 2026 First Half International Exchange Grant, Tateisi Science and Technology Foundation, Budget Amount (JPY 699,000). **--- Principal Investigator**
- 2025-2026, "Legal–Technology–Market" Collaborative Governance Model for Caring for Disabled Elderly, Zhejiang Provincial Civil Affairs Policy and Theory Research Project. **--- Co-Investigator**
- 2025-2026, Study on Institutional Innovation and Legal Safeguards for Grid-Based Smart Elderly Care in Zhejiang Province, Zhejiang Provincial Soft Science Research Program (General Project), Budget Amount (CNY 50,000). **--- Co-Investigator**
- 2024-2029, Future-Predictive Digital System by Hyper-Dimensional State Engineering, JST (Japan Science and Technology Agency), Moonshot Research and Development Program, Budget Amount (JPY 2,000,000,000). **--- Participant**
- 2024-2027, Research and Development of Technology for Monitoring Physical and Mental Health Status via Non-contact Sensing, JSPS (The Japan Society for the Promotion of Science), Grant-in-Aid for Scientific Research (C), Budget Amount (JPY 4,550,000). **--- Co-Investigator**
- 2023-2028, Bias Mitigation for Deep Neural Networks by Concept-based Image Descriptors, JSPS (The Japan Society for the Promotion of Science), Grant-in-Aid for Scientific Research (A), Budget Amount (JPY 46,800,000). **--- Participant**
- 2020-2025, Aerial-Terrestrial-Aquatic Robots for Search and Rescue in an ATA Extreme Environment, JSPS (The Japan Society for the Promotion of Science), Fund for the Promotion of Joint International Research (Fostering Joint International Research (B)), Budget Amount (JPY 18,720,000). **--- Participant**

Research Activities:

- **Reviewer Service:**
  NeurIPS, AAAI, CVPR, ICCV, ECCV, ACMMM, PR, TCSVT, WACV

Skills:

- **Models**
  LLMs, Multimodal model, Reinforcement learning, GANs, Transformer, Diffusion model.
- **Programming**
  Pytorch, Python, ROS, C/C++, Java, Android, QT, Halcon.

Languages:

- **English:** TOEIC, CET-6;
- **Japanese:** N2.